Mining Media Data I
Winter Semester 26/27
Lecture-1
Introduction

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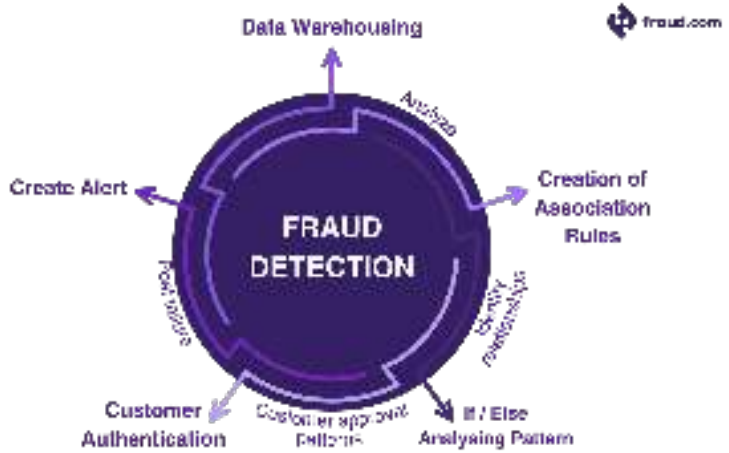
Agenda

- What is Mining Media Data (MMD)?
- Scope, literature and videos
- Organization
- Data mining in a nutshell
- Data mining tasks

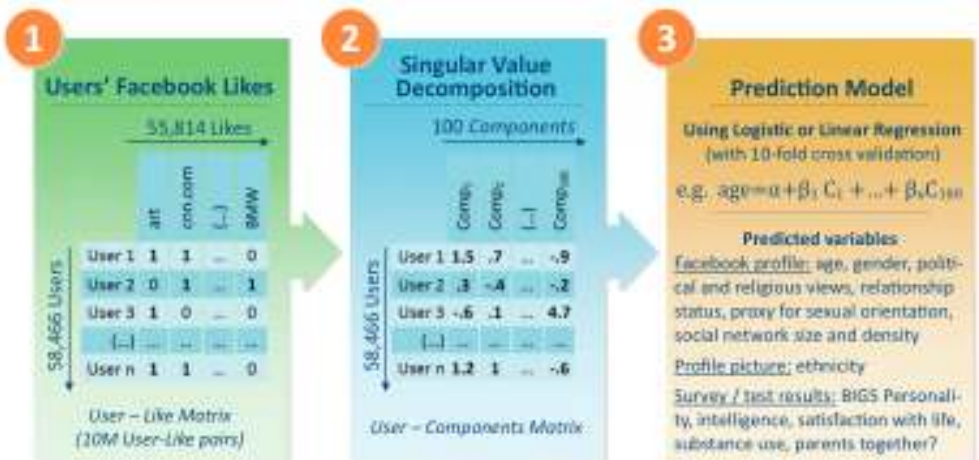
On Mining Media Data (I): Examples



Recommender systems
→ Latent Pattern Mining



Fraud detection
→ Outlier Analysis



Personality from Facebook likes (Kosinski et al., 2013) → Predictive Data Mining

On Mining Media Data (II): Applications

- Media data: user behavior, transactions, ratings, texts, images, ...
- Application areas:
  - Digital marketing and advertising
  - Content design (games, streaming, news)
  - Recommender systems
  - Recruitment
  - Digital forensics
  - Toxicity and fraud analysis
  - Resource optimization
  - Trend analysis

Scope (1): Goals

- **Established and recent** data mining methods for media data
- **Theory and practice**: preparation for work as a data analyst or data scientist in industry or research
- Our focus:
  - **mathematical optimization**
  - **representation learning**
  - **model interpretability**
  - **applications**

Scope (2): Course Structure

Block	Lectures	Guiding question	Methods
Affinity Mining	2–3	Which items occur together?	Apriori, association rules, high-utility itemsets
Latent Pattern Mining	4–6	Which hidden factors explain the data?	Recommender systems, matrix factorization, *k*-means, archetypal analysis, gradient descent, neural networks
Outlier Analysis	7	Which observations are abnormal?	Reconstruction-based methods, autoencoders
Predictive Data Mining	8–10	Which label or value applies?	Decision trees, SMOTE, digital footprints, spiking neural networks

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Scope (3): Affinity Mining

- Relationships among a predefined set of objects
- Data: *transactions*, e.g. shopping baskets
- Methods: frequent itemsets, association rules
- Applications: digital marketing, content design

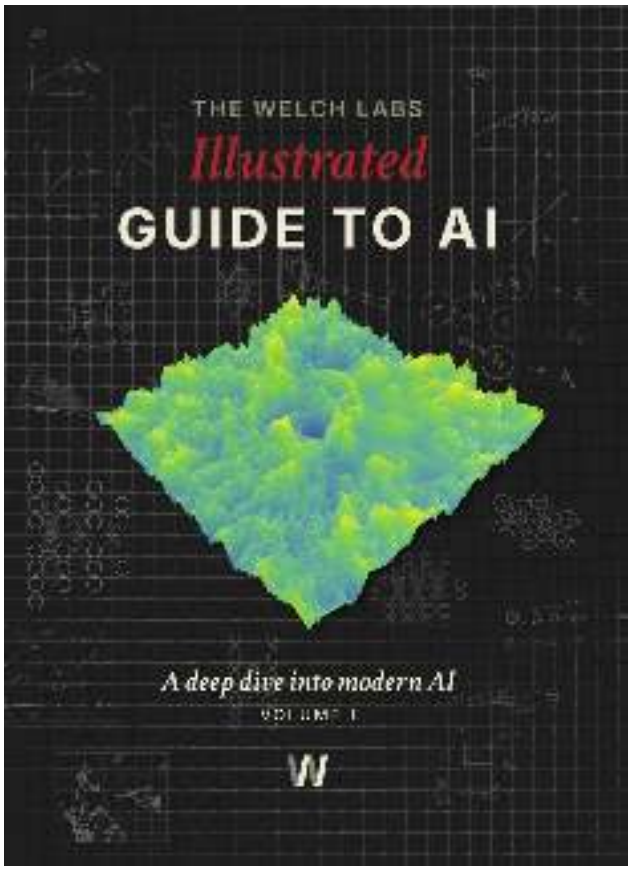
Scope (4): Latent Pattern Mining

- *Hidden* (latent) patterns in the data
- Data represented by a few automatically extracted factors
- Methods: matrix factorization, clustering, optimization
- Applications: recommender systems, behavioral profiling, digital marketing

Scope (5): Outlier Analysis and Predictive Data Mining

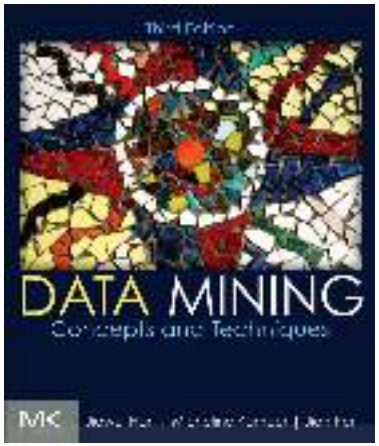
- Learning from (annotated) data to detect and predict patterns
- Methods: neural networks, autoencoders, decision trees
- Applications: fraud detection, churn prediction, user profiling

Recommended Reading (1)



- *The Welch Labs Illustrated Guide to AI: A deep dive into modern AI (Vol. I)*. Welch Labs
- **Intuition first**: core ideas of modern AI with many illustrations
- Especially for optimization and neural networks

Recommended Reading (2): More Textbooks



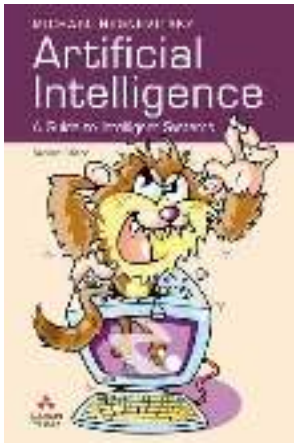
Data Mining: Concepts and Techniques (3rd Ed.)
J. Han, J. Pei, M. Kamber

Affinity Mining



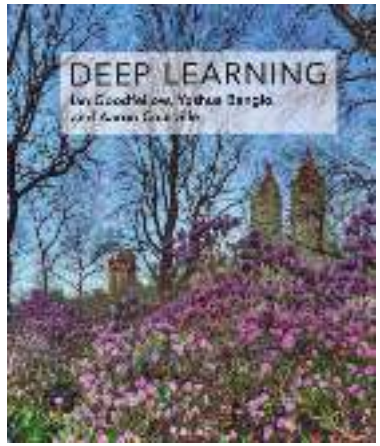
Matrix and Tensor Factorization for Profiling Player Behavior
R. Sifa

Latent Pattern Mining



Artificial Intelligence: A Guide to Intelligent Systems
M. Negnevitsky

Predictive Data Mining



Deep Learning
I. Goodfellow, Y. Bengio, A. Courville

Outlier Analysis

Recommended Videos

- **3Blue1Brown**: `https://www.youtube.com/@3blue1brown`
  - Animated mathematics: what formulas actually *mean*
  - Series *Essence of Linear Algebra* → Latent Pattern Mining
  - Series *Neural Networks* → gradient descent, backpropagation
- **Welch Labs**: `https://www.youtube.com/@WelchLabs`
  - Visual deep dives into machine learning and AI
  - The channel behind the *Illustrated Guide to AI*

Organization (1): Lecture

- **Lecture**: every Wednesday, 16:00–17:30
  - Lecture hall 1.002 (lecture hall X, main building), Nußallee 9, Bonn
- **Instructors**: Prof. Dr. Rafet Sifa, Dr. Lorenz Sparrenberg and Muskaan Chopra
- **Credits**: 4 ECTS
- **Website** (slides, assignments, news): `https://appliedmachinelearning-lab.github.io/teaching/mmdi26/`
- **Password**: IlyaSutskever
- **Questions**: `sparrenberg@bit.uni-bonn.de`

Organization (2): Schedule

Session	Date	Content	Assignment
Lecture 1	Wed, 21 Oct	Welcome / Organization	
Lecture 2	Wed, 28 Oct	Affinity Mining I	
Lecture 3	Wed, 04 Nov	Affinity Mining II	
Lecture 4	Wed, 11 Nov	Latent Pattern Mining I	
Lecture 5	Wed, 18 Nov	Latent Pattern Mining II	**Assignment 1**
Lecture 6	Wed, 25 Nov	Latent Pattern Mining III	
	Wed, 02 Dec	*No lecture*	
Lecture 7	Wed, 09 Dec	Outlier Analysis	
Lecture 8	Wed, 16 Dec	Predictive Data Mining I	**Assignment 2**
	23 Dec–06 Jan	*No lecture*	
Lecture 9	Wed, 13 Jan	Predictive Data Mining II	
Lecture 10	Wed, 20 Jan	Predictive Data Mining III	**Assignment 3**
Lecture 11	Wed, 27 Jan	Recap	
Exam 1	Wed, 24 Feb	First take	
Exam 2	Wed, 31 Mar	Second take	

Organization (3): Assignments

- **Three assignments**, presented and discussed on 18 Nov, 16 Dec and 20 Jan
  - Wednesdays, 14:00–15:30, lecture hall 1.002 (lecture hall X, main building)
- **Deadline**: day before the presentation, 23:59 (CET)
- **Groups**: up to 5 students
- **Group registration** required before the first submission

Organization (4): Group registration

- Email to `mchopra@uni-bonn.de` with
  - **group name**
  - for **each** member: first name, last name, matriculation number, Uni-ID

Example

Subject: MMDI group registration: The Apriori Avengers

First name	**Last name**	**Matriculation no.**	**Mail**
Ada	Lovelace	1234567	s0alovel@uni-bonn.de
Alan	Turing	7654321	s0aturin@uni-bonn.de

Organization (5): Assignments and LLMs

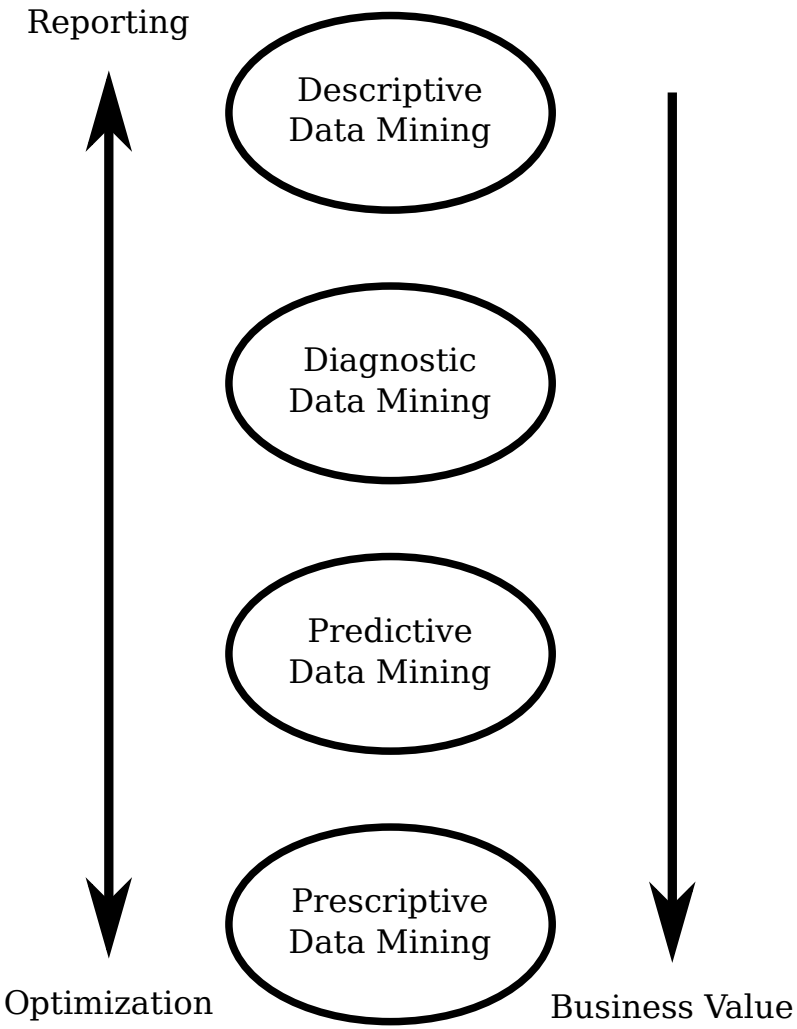
- **No rules** on using LLMs (ChatGPT, Claude, Gemini, . . .): we do not and cannot make any
- LLMs solve the assignments easily
- Learning is **your own responsibility**
- Our experience: students who solve the assignments on their own or think them through with an LLM
  - get further in the end
  - get **better grades**

Organization (6): Exam

- **Admission**: more than 50 % of the assignment points
- **Exam 1**: Wed, 24 Feb 2027
- **Exam 2**: Wed, 31 Mar 2027
- Time and room: announced during the semester

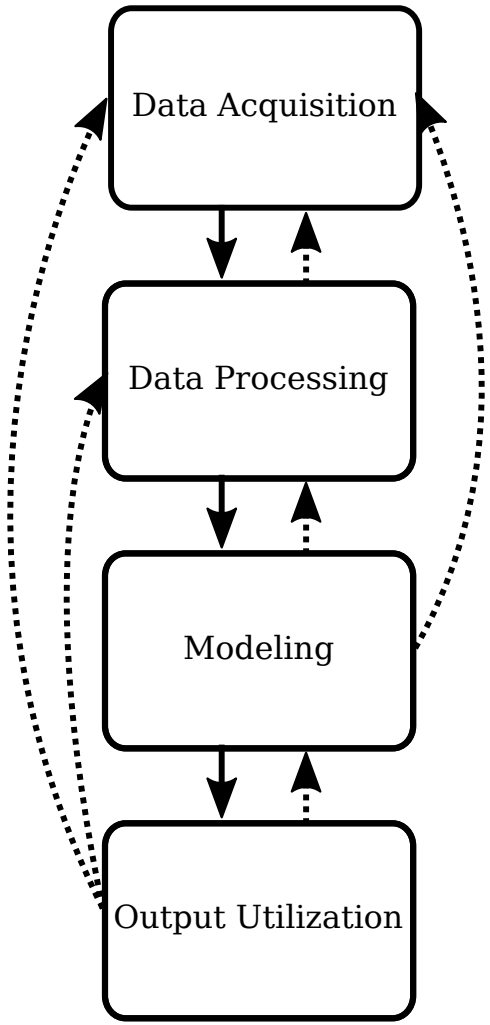
Data Mining: Taxonomy of Applications

- Data mining: modeling and extracting patterns from data
- Four types of applications:
  - **descriptive**: What happened?
  - **diagnostic**: Why did it happen?
  - **predictive**: What is likely to happen next?
  - **prescriptive**: What should we do about this?



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Data Mining: Workflow



- **Data Acquisition**: collect raw data (logs, transactions, ratings, texts)
- **Data Processing**: clean, filter, extract features
- **Modeling**: apply a data mining method, e.g. clustering or classification
- **Output Utilization**: interpret, visualize and use the results
- **Dotted arrows**: feedback loops, the process is *iterative*

Data Mining: Main Challenges

- **Multichannel input**: many sources (clicks, transactions, texts, images, . . .) to combine
- **Minority**: rare events, pattern or outlier?
- **Scalability**: millions of users and items
- **Interpretability for interoperability**: results must be understandable for those who act on them
- **Data dependency**: results are only as good as the data (quality, bias, changing behavior)

Data Mining Tasks (1): Affinity Mining

Customers who bought the item(s):



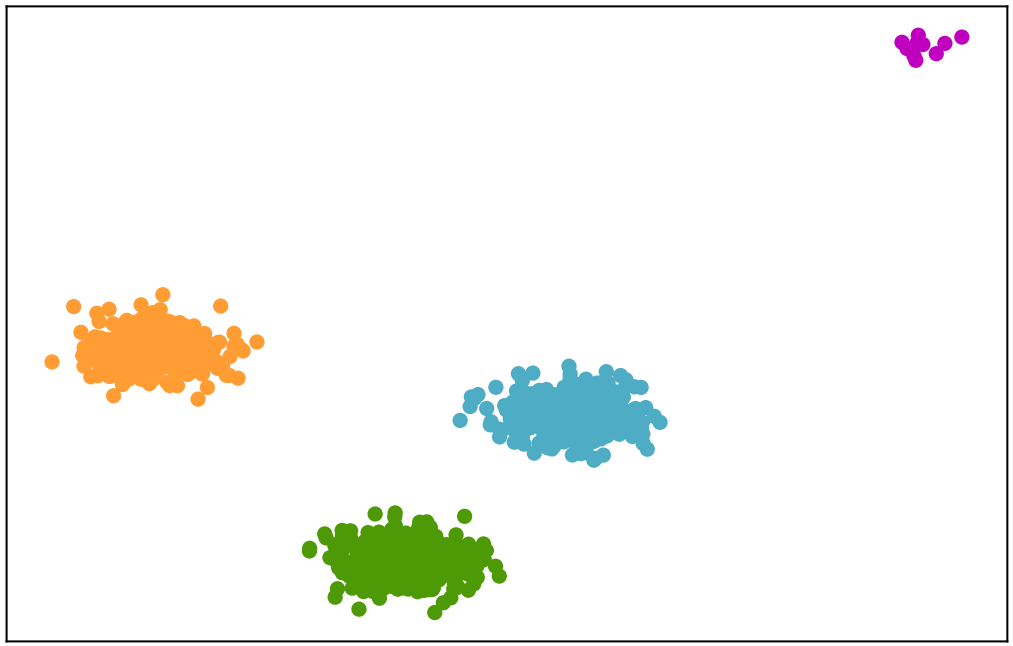


Also bought :



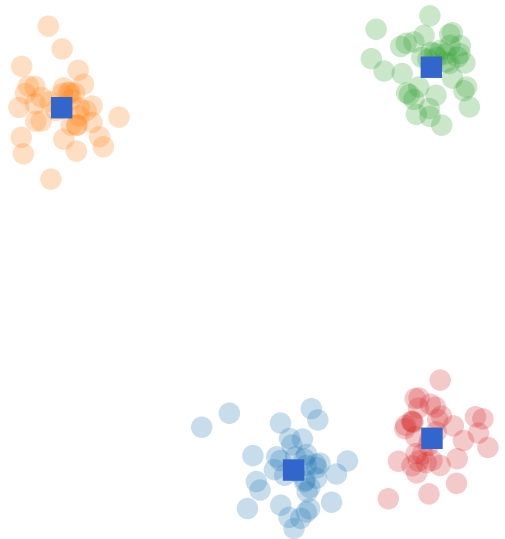
- Relationships among entities, e.g. items bought together
- Automatically extracts interesting local patterns
- Used in descriptive, diagnostic and prescriptive data mining

Data Mining Tasks (2): Outlier Detection



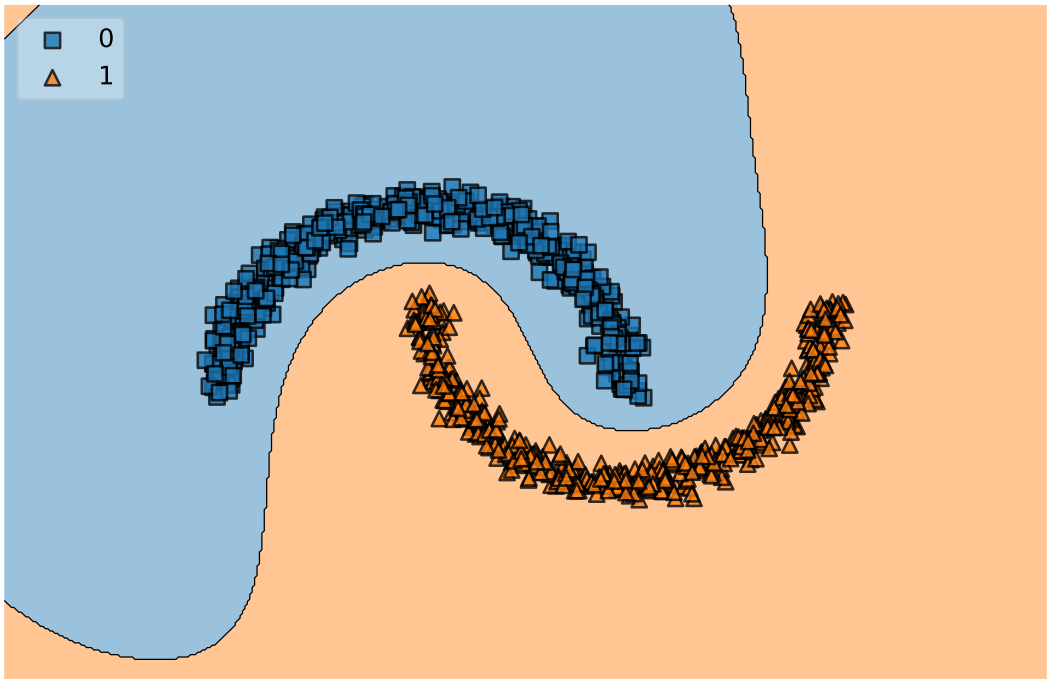
- Identifying *abnormal* observations
- E.g. fraud detection

Data Mining Tasks (3): Clustering



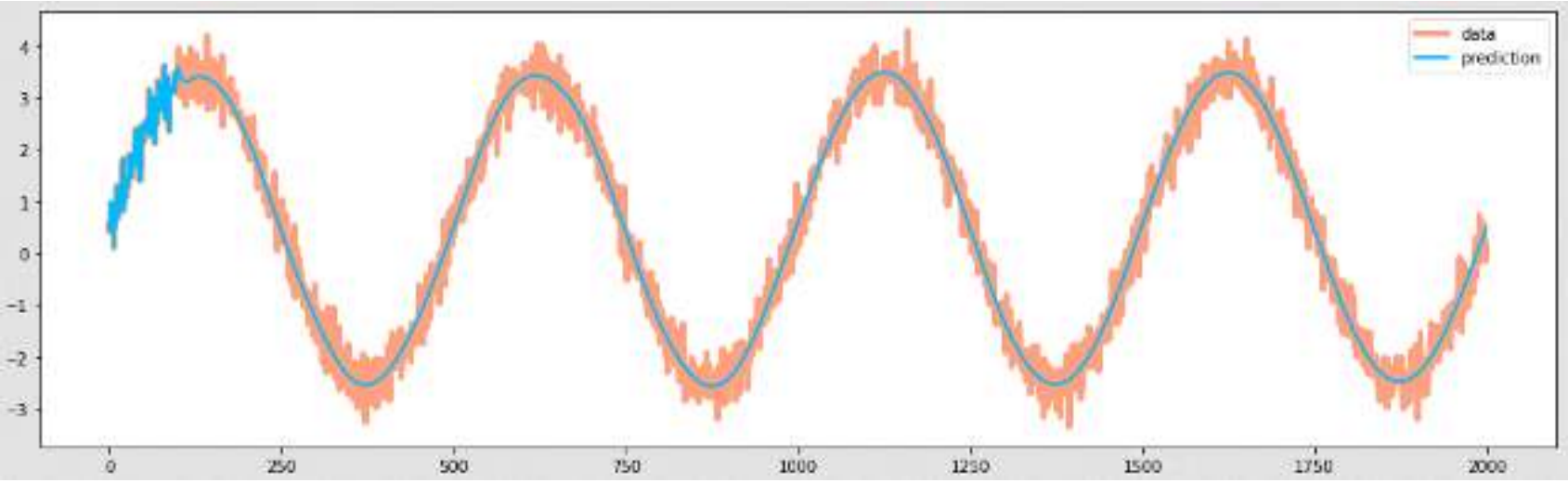
- Identifying similar observations
- No predefined *classes*
- E.g. behavioral profiling

Data Mining Tasks (4): Classification



- Predicting a predefined label
- E.g. churn prediction: will a player of a free-to-play game quit?

Data Mining Tasks (5): Regression



- Predicting a numerical value
- E.g. customer lifetime value: how much will a customer spend?

Next Lecture and Assignment

- **Next lecture** (28 Oct): Affinity Mining I, frequent itemsets and the Apriori algorithm
- **Assignment 1**
  - Published on the website in the next few days: `https://appliedmachinelearning-lab.github.io/teaching/mmdi26/`
  - Password: IlyaSutskever
  - Deadline: **17 Nov 2026, 23:59 (CET)**
  - Presentation: 18 Nov 2026
  - Submit to `mchopra@uni-bonn.de`, subject: group name and assignment number